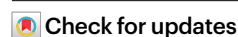


White matter pathways mediating dorsolateral prefrontal TMS therapy for depression

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We use connectome modeling to map polysynaptic fiber pathways underlying transcranial magnetic stimulation (TMS) therapy for depression. We propose putative cortical and subcortical routes that connect stimulation sites in the dorsolateral prefrontal cortex to the subgenual cingulate cortex via intermediate regions. Here we show that route length explains both TMS treatment response in two independent patient cohorts and the clinical efficacy of functional magnetic resonance imaging-guided TMS targeting. Our results illuminate the neuroanatomical basis of TMS therapy for depression.

Transcranial magnetic stimulation (TMS) of the left dorsolateral prefrontal cortex (DLPFC) is a therapy for refractory depression¹. The therapy's efficacy is thought to relate to indirect modulation of the subgenual cingulate cortex (SGC)—a structure that is not directly accessible to TMS. Support for this idea comes from observations that stimulating the DLPFC elicits acute and lasting changes in SGC activity and connectivity, as demonstrated by TMS combined with functional magnetic resonance imaging (fMRI)^{2,3} and intracranial recordings⁴. Moreover, multiple fMRI studies have indicated that baseline functional connectivity between DLPFC TMS sites and the SGC predicts patient treatment responses^{5–7}.

Interestingly, there is no evidence for strong axonal projections from the DLPFC to the SGC. Human brain imaging has not identified white matter tracts that could establish a direct communication channel

between the two regions⁸. Tract-tracing work in rhesus monkeys has reported only sparse monosynaptic projections from the DLPFC to the SGC^{9,10}, instead arguing for an indirect route via pregenual area 32 (ref. 11). The propagation of DLPFC stimulation to the SGC is therefore hypothesized to be polysynaptic, that is, mediated by a sequence of intermediate regions interconnected via white matter fibers. Current methods to map white matter circuits using brain stimulation are limited to direct, 'one-hop' fibers between anatomically connected regions¹², making them unsuitable for characterizing DLPFC–SGC communication¹³. As a result, despite the progress achieved using fMRI, the neuroanatomical circuits mediating the therapeutic outcomes of TMS in depression remain elusive. This knowledge gap poses a central barrier to understanding and leveraging the role of the human structural connectome in mediating the response to TMS therapy.

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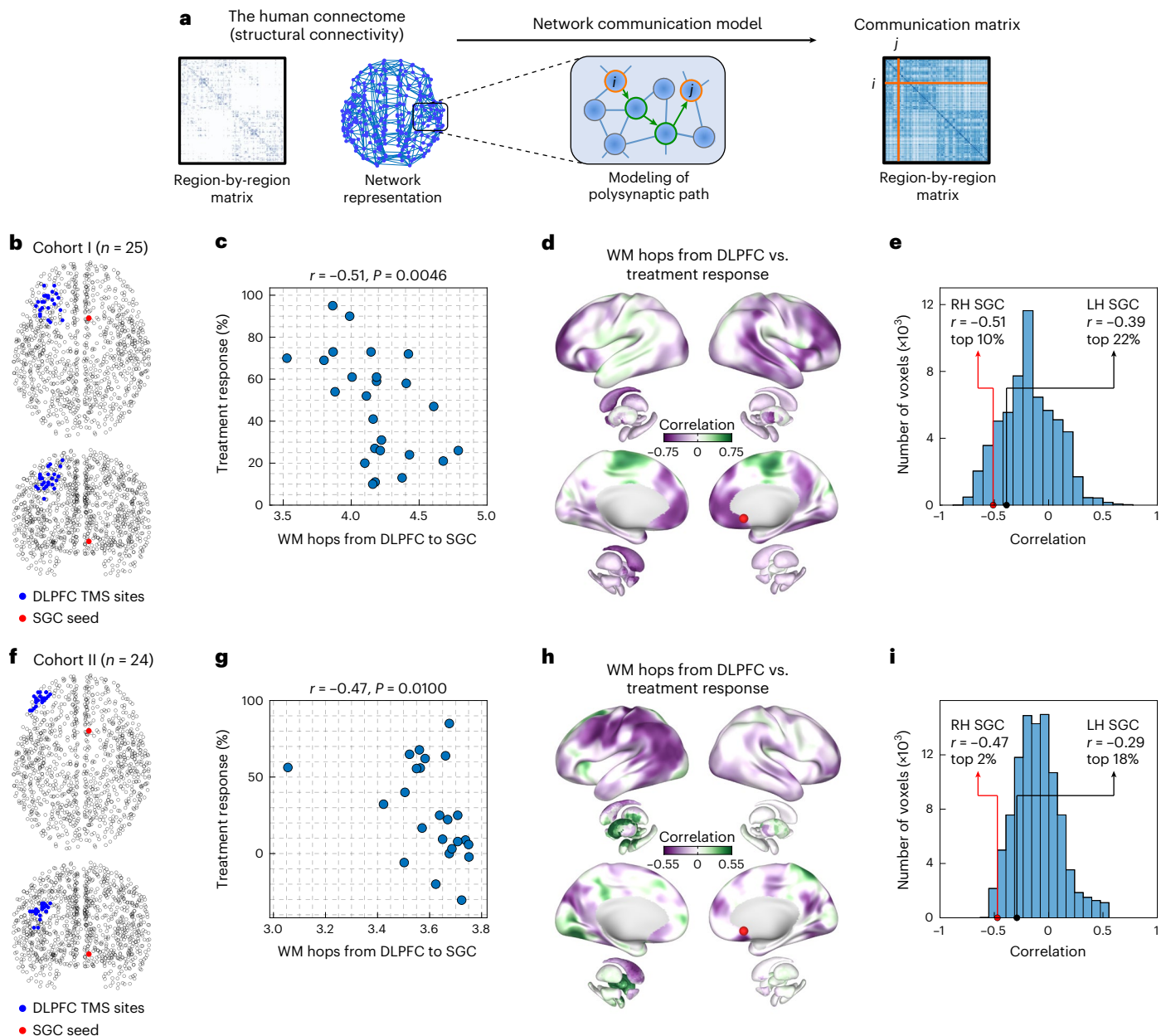


Fig. 1 | Association between DLPFC–SGC white matter pathways and TMS treatment response. **a**, Network communication models quantify the capacity for signaling via the structural connectome. The models identify putative paths of polysynaptic communication between regions not connected by a direct fiber. **b**, TMS and SGC coordinates in cohort I. Black circles denote the gray matter regions of the connectome. **c**, Relationship between treatment response and the number of white matter tracts traversed (white matter (WM) hops) from DLPFC stimulation sites to the SGC in cohort I. Patients stimulated at DLPFC sites with shorter polysynaptic paths to the SGC showed better treatment outcomes (Spearman rank correlation coefficient $r = -0.51$, one-tailed $P = 0.0046$).

d, Cortical map of the correlation between white matter hops from DLPFC stimulation sites and treatment response in cohort I. Downstream regions for which efficient communication from the DLPFC was associated with better (worse) clinical outcomes are shown in purple (green). Treatment response was prominently associated with fewer white matter hops from the DLPFC stimulation site to the right SGC (red dot, MNI (6, 16, -10)), while also highlighting other regions of potential interest. **e**, Distribution of the correlations shown in **d**. Voxels within TMS activation areas were excluded from the histogram. RH, right hemisphere; LH, left hemisphere. **f–i**, Same as **b–e** but in cohort II (in **g**, Spearman rank correlation coefficient $r = -0.47$, one-tailed $P = 0.0100$).

Here, we used advances in connectome modeling to map polysynaptic fiber pathways from the DLPFC to the SGC and show that they explain TMS antidepressant effects. Network communication models quantify the capacity of signal transmission between regions via the connectome (Fig. 1a)^{14,15}. The models can identify multihop pathways (sequences of two or more white matter tracts) between gray matter structures that are not connected by a direct fiber. We used this modeling framework to reveal putative neuroanatomical routes traversed by TMS in two cohorts of individuals with depression.

We considered two independent cohorts of individuals with treatment-resistant depression who underwent repetitive TMS to the left DLPFC (cohort I: $n = 25$, 17 females, mean $\pm$ s.d. age of 54.8 ± 9.9 years⁶; cohort II: $n = 24$, 10 females, 44.5 ± 11.8 years⁷; Methods). Stimulation sites in the DLPFC were determined separately for each patient using established methods (Fig. 1b,f). Treatment response was measured as the percentage change in the Montgomery–Åsberg Depression Rating Scale (MADRS; cohort I) and Beck Depression Inventory (BDI; cohort II) scores before and after TMS therapy. We used network

communication models to map polysynaptic fiber pathways from volumes of activated tissue centered at TMS coordinates in the left DLPFC to the right SGC (10-mm-radius sphere centered at MNI (6, 16, -10), as per refs. 5–7). We evaluated nine established network measures spanning different conceptualizations of how signals propagate through the connectome. We focused on pathways mapped using the shortest path routing model, which showed the most consistent associations across patient cohorts. In this model, a pathway's capacity to support TMS propagation was measured as its average number of hops through the connectome (that is, the number of white matter tracts traversed from the stimulation site to the SGC). Pathways were mapped on normative connectomes, constructed with the Lausanne (discovery) and Schaefer (replication) parcellations (Methods).

We found that polysynaptic white matter pathways were associated with treatment response in both cohorts. Patients stimulated at DLPFC sites connected to the SGC by fewer white matter hops—shorter, more direct communication channels—showed higher rates of clinical improvement (Lausanne connectome, cohort I: Spearman rank correlation coefficient $r = -0.51$, one-tailed $P = 0.0046$, Fig. 1c; cohort II: $r = -0.47$, $P = 0.0100$, Fig. 1g). This suggests that white matter connectivity has a key role in polysynaptic signaling from precise DLPFC stimulation sites to the SGC, and that variations in TMS coordinates—leading to differences in how efficiently signals are transmitted via the connectome—contribute to heterogeneity in patient outcomes.

To assess the robustness of this finding, we recomputed our analysis using an alternative connectome-mapping pipeline, comprising different image acquisition and preprocessing sequences, tractography algorithm and gray matter parcellation. Once again, we found that patients stimulated at DLPFC sites with shorter polysynaptic paths to the SGC showed better treatment outcomes (Supplementary Fig. 1; Schaefer connectome, cohort I: $r = -0.54$, $P = 0.0026$; Schaefer connectome, cohort II: $r = -0.41$, $P = 0.0223$). These associations were robust to leave-one-out outlier sensitivity tests (Supplementary Fig. 2). Of the network measures evaluated, the number of steps along the shortest path was the only measure associated with treatment outcome across both patient cohorts and connectome-mapping pipelines (Supplementary Fig. 3; correlations were significant in two of the four scenarios following correction for multiple comparisons across nine network measures: Lausanne connectome, cohort I: false discovery rate Benjamini–Hochberg adjusted $P = 0.0205$; Lausanne connectome, cohort II: $P = 0.0897$; Schaefer connectome, cohort I: $P = 0.0205$; Schaefer connectome, cohort II: $P = 0.20$).

To further investigate this, we constructed surrogate connectomes by rewiring connections between brain regions. Surrogate connectomes are brain-like networks that preserve key features of white matter connectivity (here, the number of tracts intersecting each region, the tract length distribution, and the relationship between tract weight and length) but are otherwise randomly shuffled into an alternative realization of brain wiring¹⁶. We observed a significant decrease in the association between fiber communication and treatment response in surrogate connectomes (Supplementary Fig. 4), underscoring the importance of the biological topography of human white matter fibers for TMS propagation and its therapeutic efficacy in depression.

While there is accumulating evidence implicating the right SGC in TMS therapy for depression^{2–7}, could modulation of other downstream regions be more relevant to treatment response? To investigate this, we computed white matter pathways not only to the right SGC but also to 10-mm spheres centered at each gray matter voxel in the brain, yielding a whole-brain map of the correlation between TMS propagation and treatment response (Fig. 1d,h). Our analysis corroborates previous reports on the importance of the SGC seed centered at MNI (6, 16, -10)^{5–7}, with this coordinate positioned among the top 10.17% and top 2.25% of voxels most strongly associated with patient outcomes in cohorts I and II, respectively (Fig. 1e,i, Lausanne connectome; Supplementary Fig. 1c,f, Schaefer connectome). Other prominent regions included portions of

the medial superior, orbital frontal and superior temporal cortices, as well as the precuneus and caudate. Interestingly, the contralateral left SGC seed (MNI (6, 16, 10)) yielded weaker correlations to treatment response, suggesting a lateralized role of this structure consistent with previous work^{5,6}. We also noted the presence of a few regions for which longer communication pathways were associated with better treatment response, suggesting that dampening TMS modulation of certain areas may also contribute to treatment efficacy.

Our analyses to this point presupposed a serial pathway of activation linking the DLPFC to the SGC. An alternative possibility is that TMS therapy is mediated by the parallel engagement of distributed downstream regions. For example, deep brain stimulation targets for depression often co-activate neighboring structures within the cingulo-striatal circuit^{17,18}. To test this, instead of the right SGC sphere, we repeated our analysis using a bilateral circuit mask encompassing the SGC, nucleus accumbens and caudate. Across both cohorts and connectome reconstructions, correlations with treatment response were comparable or stronger for the right SGC alone relative to the expanded circuit (Supplementary Fig. 5), supporting the specificity of the SGC as a key downstream target of DLPFC stimulation.

What neuroanatomical structures underpin TMS communication from the left DLPFC to the right SGC? To answer this question, we combined modeling results across individuals of the same cohort and counted the occurrences of the network paths used to infer white matter communication (Fig. 2a,d). In agreement with evidence for a polysynaptic DLPFC–SGC route^{8,11}, we found that all paths were formed by a sequence of at least two steps, that is, they traversed at least one intermediate region (Fig. 2b,e). Paths were consistent across both cohorts and mediated by key structures such as the thalamus, dorsal and medial areas of the superior frontal gyrus (SFG), and the anterior cingulate cortex (Fig. 2c,f). The paths clustered into three main neuroanatomical routes—one cortical pathway and two complementary fronto-thalamic circuits (Fig. 2h). The cortical route comprised three polysynaptic steps: from stimulation sites in the left DLPFC to the right dorsal SFG via the corpus callosum, to the right anterior cingulate cortex and to the right SGC via the cingulum bundle (Fig. 2g,h, blue arrows). Both fronto-thalamic routes comprised four steps and differed primarily in their interhemispheric crossing point. The first and more frequently observed route proceeded from the left DLPFC to the left thalamus, then to the left medial SFG, crossed homotopically to the right SFG and finally reached the right SGC (Fig. 2h, green arrows). The second fronto-thalamic route crossed hemispheres earlier, from the left DLPFC to the right dorsal SFG, to the right thalamus, to the right medial SFG and to the right SGC (Fig. 2h, orange arrows). We also noted the presence of a minor ventral limbic pathway via the right thalamus, amygdala and temporal pole. Supplementary Figs. 6–15 show the top five pathways from each cohort in MNI volumetric space.

How do these neuroanatomical pathways advance previous work on fMRI-guided TMS for depression? Prior evidence has suggested that TMS is more effective when targeted to DLPFC sites with strong negative functional connectivity (that is, anticorrelation) to the SGC^{5–7} (although see ref. 2). As an initial check, we first confirmed this association in both cohorts of individuals with depression using functional connectivity from the normative connectome datasets (Supplementary Fig. 3; Lausanne connectome, cohort I: $r = -0.46$, $P = 0.0102$; Lausanne connectome, cohort II: $r = -0.37$, $P = 0.0366$; Schaefer connectome, cohort I: $r = -0.48$, $P = 0.0079$; Schaefer connectome, cohort II: $r = -0.35$, $P = 0.045$). Although structural communication yielded stronger associations with treatment outcome than functional connectivity, this difference was not statistically significant (bootstrap nonparametric $P > 0.05$ for both cohorts and connectome datasets). Adding both predictors in a bivariate model did not improve explanatory power (all F -test $P > 0.05$), suggesting that structural communication and functional connectivity explain overlapping variance in treatment outcome.

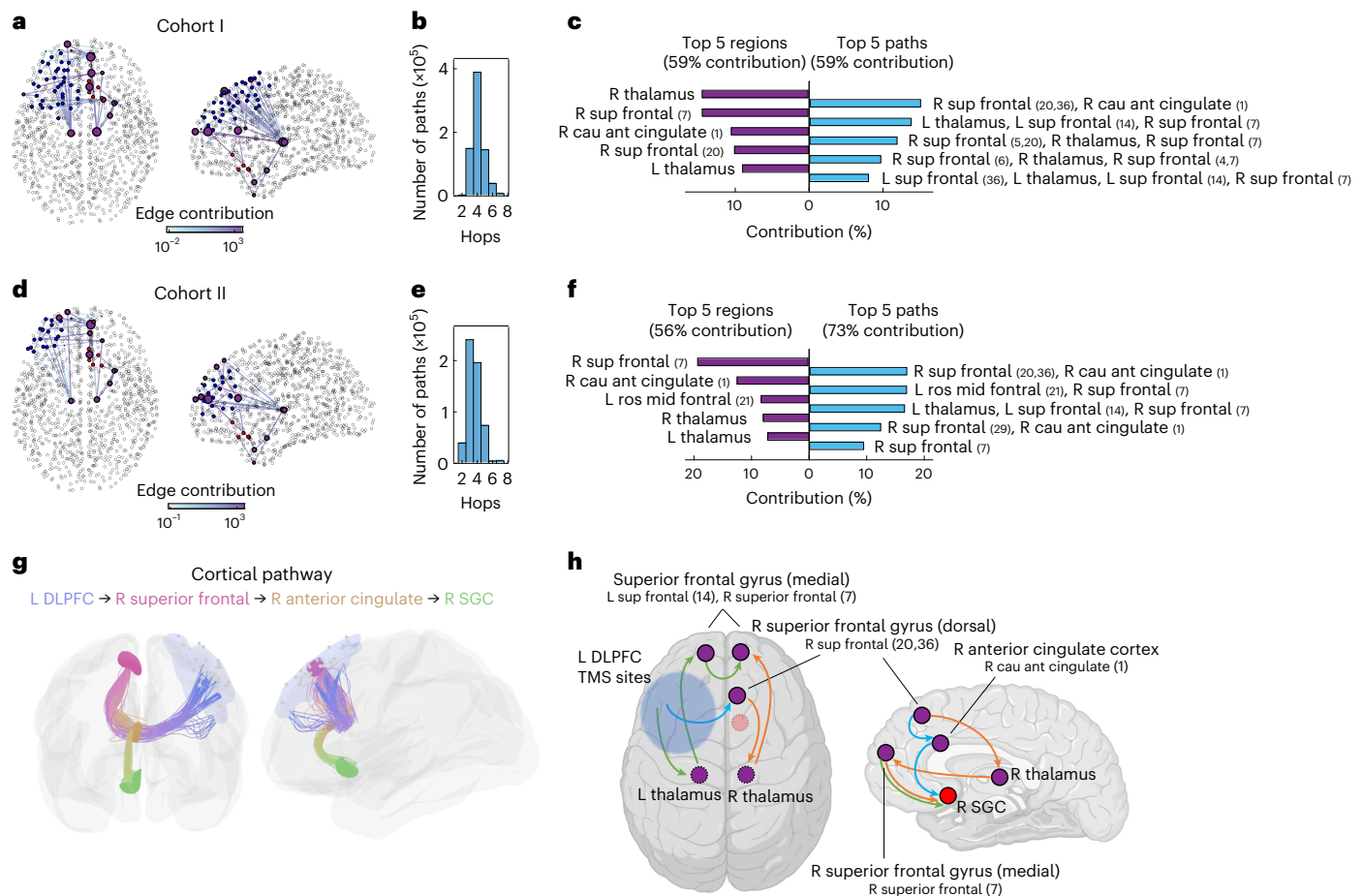


Fig. 2 | Neuroanatomical DLPFC-SGC pathways. **a**, Network visualization of paths originating from stimulated gray matter regions in the left DLPFC (blue), via intermediate regions (purple), to right SGC regions (red) in cohort I. The size of intermediate regions and the color of connections are proportional to the number of times they are traversed. **b**, Number of white matter hops from stimulated to SGC regions in cohort I. **c**, Top five intermediate regions and paths in cohort I. Spatially adjacent gray matter parcels were grouped into the same parcellation labels for clarity (for example, R sup frontal 20 and 36). **d–f**, Same

as **a–c** but in cohort II. **g**, Anatomical visualizations of the cortical pathway from the left DLPFC to the right SGC reconstructed using tractography. Fiber colors show different white matter tracts along the connectome. **h**, Schematic of the three main neuroanatomical routes from the left DLPFC to the right SGC. When relevant, the regions are described with both anatomical nomenclature and labels from the Lausanne parcellation. The positions of the regions are approximate (see Supplementary Figs. 6–15 for detailed MNI coordinates). Brain illustrations in **h** created in BioRender; Seguin, C. <https://biorender.com/qt313ao> (2026).

Next, we computed two modalities of DLPFC maps: the first based on the functional connectivity to the SGC and the second based on the number of hops via white matter pathways to the SGC (Fig. 3). We found a positive correlation between the two maps, indicating that DLPFC sites with negative functional connectivity to the SGC also tend to be connected to it via shorter, more direct polysynaptic white matter pathways (Fig. 3c,f; Lausanne connectome, $r = 0.48$; Schaefer connectome, $r = 0.75$; both $P < 10^{-12}$). Lastly, we computed the DLPFC coordinates that minimize the polysynaptic path length to the SGC (Fig. 3a,d; Lausanne connectome, MNI (−44, 52, 22); Schaefer connectome, MNI (−44, 54, 16)) and observed a negative correlation between the distance to these coordinates and clinical improvement, suggesting that stimulation nearer to this model-derived reference point tended to yield better outcomes (Supplementary Fig. 16). Together, these findings provide a neuroanatomical basis for the efficacy of fMRI-guided TMS targeting in depression and position connectome-based models as a promising framework for refining stimulation strategies.

In sum, we used network communication models to illuminate the neuroanatomical basis of DLPFC-targeted TMS for depression. We show that structurally constrained communication from stimulation sites at the left DLPFC to the right SGC is correlated with patient outcomes. While the sample sizes in our analyses were modest, we replicated our results in two independent cohorts of individuals with depression and

two independent normative connectomes, demonstrating generalizability across different patient populations, TMS protocols, neuroimaging sequences and connectome-mapping methods.

We propose cortical and subcortical circuits involved in the therapeutic mechanisms of TMS for depression. The subcortical pathway adds support to literature implicating fronto-thalamic networks in depression¹⁹, while the cortical pathway mediated by the anterior cingulate cortex (Brodmann area 32) recapitulates tract-tracing findings in monkeys of a DLPFC-SGC route via the pregenual area 32 (refs. 10,11). Importantly, these pathways remain putative and are susceptible to the inherent limitations of diffusion MRI and tractography. For example, the cortical pathway progressed from the left DLPFC to the right medial SFG—a heterotopic connection that may reflect a tractography bias to terminate interhemispheric tracts along the medial wall²⁰. Related to this, cortico-thalamo-cortical circuits often involve relays via striatal and pallidal nuclei²¹, but difficulties in tracking short-range fibers within the basal ganglia may have downplayed the influence of these canonical loops in our models. Future work is therefore necessary to validate our findings, for example, by using intracranial recordings to track evoked responses along the proposed routes following DLPFC stimulation.

Our results establish the utility of structural connectivity—when coupled with computational models of neural communication^{22–24}—to

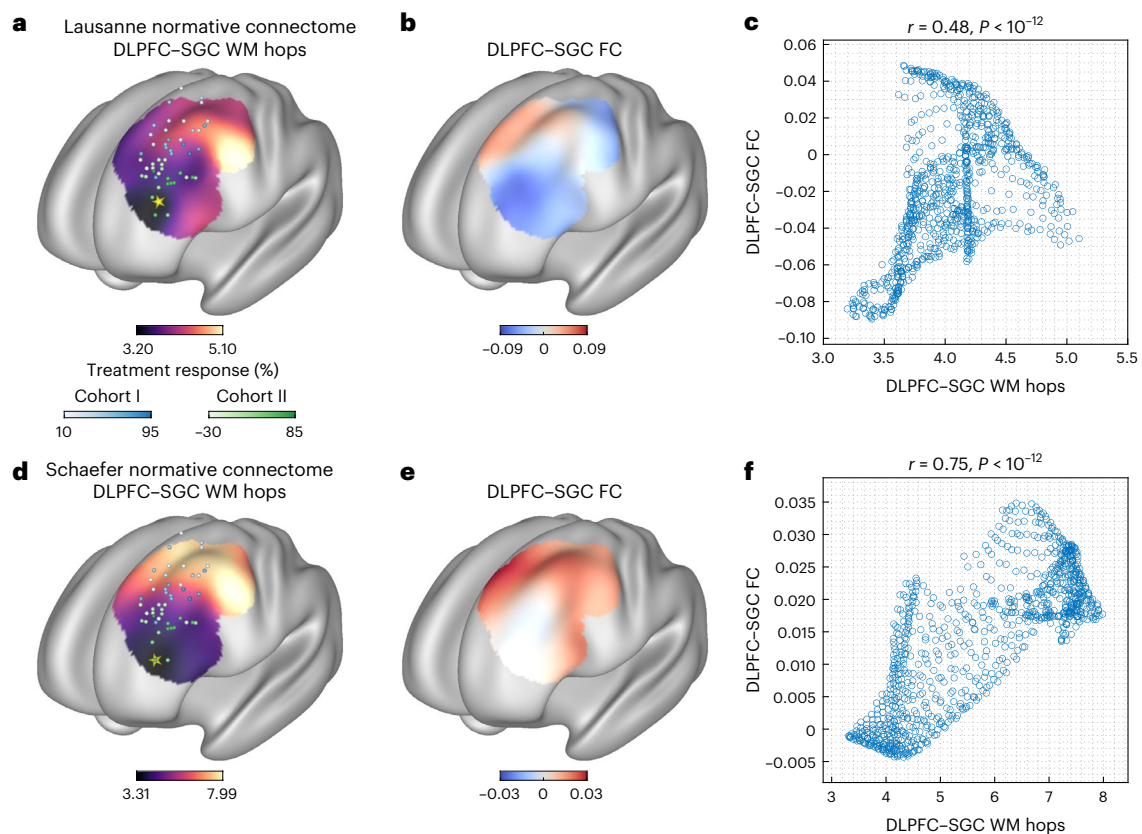


Fig. 3 | Structural and functional DLPFC–SGC maps. a,b,d,e, Maps showing the number of white matter hops to the SGC (**a,d**) and the functional connectivity (FC) to the SGC (**b,e**) in two normative connectome datasets. Spheres mark TMS sites in cohorts I (blue) and II (green) and are colored according to the patient's treatment response. Gold stars mark the DLPFC loci that minimize the polysynaptic path length to the SGC (Lausanne, MNI (−44, 52, 22); Schaefer, MNI

(−44, 54, 16)). **c,f,** Scatterplots showing the association between DLPFC–SGC white matter communication and functional connectivity in two normative connectome datasets (two-sided Spearman rank correlation, both $P < 10^{-12}$). Each dot represents a gray matter voxel in the DLPFC that is accessible to TMS. DLPFC sites functionally anticorrelated with the SGC tend to have shorter, more direct polysynaptic fiber pathways to the SGC.

inform stimulation protocols based on polysynaptic fiber pathways. We focused on simple graph-theoretical models that estimate the capacity for inter-regional communication via the connectome. The key advantages of these models are interpretability and computational tractability, offering a straightforward method to quantify TMS propagation routes that can be readily translated into clinical interventions. In contrast, more complex biophysical models of neural dynamics may better capture the underlying neurophysiological mechanisms of TMS and will be valuable for future mechanistic work²³. Among the network measures evaluated, the number of steps along the DLPFC–SGC shortest path was the strongest predictor of treatment response. This finding contrasts with earlier evidence that diffusion-based models more accurately predict electrical stimulation spread in the human connectome²². We speculate that diffusion processes may better reflect global patterns of network activation, whereas shortest paths could be more relevant for the targeted, circuit-specific effects mediating antidepressant response. Emerging datasets capturing real-time evoked activity from interleaved TMS–fMRI will help clarify these distinctions²⁵.

More broadly, a promising prospect is the use of computational models to augment current strategies for imaging-guided stimulation. Mapping polysynaptic paths may enhance existing methods based on direct tracts, such as fiber filtering and lesion network mapping^{12,13}. We found that structural communication provides an anatomical grounding for fMRI-guided targeting in depression. Although neither imaging modality clearly outperformed the other in explaining treatment efficacy, prospective trials with larger sample sizes could explore

integrating both structural and functional approaches into hybrid targeting strategies. In the future, our work could also be extended to evaluate and inform personalized targeting based on each individual's unique connectome⁸, across brain stimulation modalities, including TMS and deep brain stimulation.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41593-026-02248-6>.

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Methods

Clinical cohorts

We considered two previously published cohorts of individuals with treatment-resistant depression who underwent repetitive TMS to the left DLPFC. Cohort I comprised 25 individuals (17 females, mean $\pm$ s.d. age of 54.8 ± 9.9 years) treated at the Berenson-Allen Center at Beth Israel Deaconess Medical Center (BIDMC), Boston, USA⁶. The study was approved by the internal review board of BIDMC, and participants provided written informed consent. Patients were diagnosed with major depression and met the criteria for medication resistance, including failure to respond to at least one antidepressant. The treatment consisted of repetitive high-frequency TMS to the left DLPFC (10 and 20 Hz at 110% and 120% of the resting motor threshold, respectively) delivered daily for a period of 4–7 weeks. Precise stimulation sites were determined by moving 5.5 cm anterior to the motor threshold location along a parasagittal line. TMS coordinates were recorded using a neuronavigation system and later transformed to MNI space using the patients' T1-weighted anatomical scans. Treatment response was measured by the change in self-reported depressive symptom scores using the BDI-II, assessed on the first and last day of therapy. The mean $\pm$ s.d. baseline BDI score of 38.6 ± 9.3 was significantly decreased to 21.2 ± 13.0 following TMS treatment ($P < 0.0001$). Complete details on the clinical characterization, data acquisition and TMS treatment for cohort I are reported in ref. 6.

Cohort II contained 24 individuals (10 females, 44.5 ± 11.8 years old) who completed 5–8 weeks of repetitive TMS to the left DLPFC as part of a larger clinical study (Australian New Zealand Clinical Trials Registry: Investigating predictors of response to transcranial magnetic stimulation for the treatment of depression; ACTRN12610001071011)⁷. The study was approved by the human research ethics committee of Alfred Hospital, and participants provided written informed consent. Individuals met the key inclusion criteria of major depression diagnosis and treatment-resistant depression at stage III of the Thase and Rush classification. Repetitive high-frequency TMS was delivered to the left DLPFC at 10 Hz and 110% of the motor threshold. Patient-specific stimulation coordinates were determined using the F3 beam approach and later transformed to MNI space using anatomical scans. Treatment response was measured as the difference in self-reported MADRS scores from baseline to 6 weeks after the start of TMS therapy. The mean $\pm$ s.d. baseline MADRS score of 33.7 ± 6.7 was significantly decreased at 6 weeks to 24.3 ± 11.4 ($P = 0.0003$). Further details on the patient demographics, data acquisition and TMS treatment for cohort II are found in ref. 7 and at <https://www.anzctr.org.au/Trial/Registration/TrialReview.aspx?id=336262>.

Datasets of normative brain connectivity

We used two previously published and openly available datasets of normative structural and functional connectivity. Normative connectomes—mapped from high-quality brain images of healthy adults—are commonly used to inform clinical brain stimulation studies as they have a higher signal-to-noise ratio than connectivity reconstructed from patient data^{12,26}. As detailed below, we used an established method to generate normative connectomes that mitigates tractography biases and filters out individual-level artifacts^{27,28}. These factors contribute to the accurate delineation of white matter fibers and pathways associated with the clinical outcomes of brain stimulation^{12,29}.

The discovery normative dataset, used in all analyses in the main text, comprised brain networks from 70 healthy young adults (mean $\pm$ s.d. age of 28.8 ± 9.1 years, 27 women) acquired at the University of Lausanne, Switzerland³⁰. Gray matter was parcellated into 1,000 cortical regions and 14 subcortical regions (left and right thalamus, caudate, putamen, pallidum, accumbens area, hippocampus and amygdala)³¹. Structural connectivity was estimated from the diffusion MRI of individual participants using deterministic white matter tractography. The strength of the white matter connection between two regions was

computed as the number of streamlines intersecting them divided by the average of their surface areas. Resting-state fMRI volumes underwent preprocessing to correct for motion, physiological confounders and global signal. The functional connectivity of each individual was computed as the Pearson correlation between regional time courses of blood-oxygenation-level-dependent (BOLD) signals. Further details on image acquisition and connectivity reconstruction are found in ref. 30.

The replication dataset comprised brain networks from 1,000 healthy young adults from the Human Connectome Project (HCP). Details on HCP image acquisition and preprocessing are described in refs. 32–34. Gray matter regions were defined based on the Schaefer 800 (7 networks) parcellations³⁵ combined with the same 14 subcortical regions listed above. Structural connectivity was computed using probabilistic white matter tractography applied to diffusion MRI, and the strength of the structural connections was inferred as the streamline count between gray matter regions. Functional connectivity was once again computed as the Pearson correlation of regional BOLD time courses following resting-state fMRI preprocessing. Further details on connectivity reconstruction are provided in ref. 36.

We applied an established consensus method to generate group-representative connectomes from the structural connectivity matrices of individual participants²⁷. The method selects the white matter connections most reliably found across individuals, while preserving the sample's average connection density, numbers of intra- and interhemispheric connections and distribution of connection lengths. This approach yields a prototypical map of human brain connectivity that mitigates tractography reconstruction errors, filters out individual-level artifacts and boosts signal-to-noise ratio^{27,28}. For the discovery dataset, this yielded a $1,014 \times 1,014$ structural connectivity matrix with 2.4% connection density. For the replication dataset, we obtained an 814×814 structural connectivity matrix with 27.8% connection density.

Neuroanatomically constrained connectomes

Structural connectivity matrices mapped using diffusion MRI and tractography are subject to well-established limitations. In particular, the presence of false-positive links can influence models of polysynaptic communication and lead to pathways that are not biologically realistic. For example, tract-tracing evidence in nonhuman primates shows that the SGC sends monosynaptic projections to subcortical nuclei, but these links are unidirectional—oriented exclusively from the SGC to the subcortex. However, because diffusion MRI cannot resolve axonal directionality, tractography-based connections between the SGC and subcortical regions are bidirectional and encoded as symmetric entries in the structural connectivity matrices described above. The difficulty of reconstructing fibers with overlapping or complex geometries, such as those traversing the basal ganglia, can also generate spurious connections that are unlikely to exist in the human brain. While the connectomes considered in our study are publicly available and have been used in numerous studies^{27,30,36}, these limitations are particularly relevant to the goal of mapping polysynaptic communication pathways.

To address these issues, we preprocessed the structural connectivity matrices to constrain their connections using tract-tracing evidence from nonhuman primates. This procedure systematically and conservatively removed links that were likely to be false positives based on established neuroanatomical knowledge. Both normative connectomes considered in this study—the Lausanne (1,014 regions) and Schaefer (814 regions) parcellations—include the same 14 FreeSurfer subcortical regions (left and right thalamus, caudate, putamen, pallidum, accumbens, hippocampus and amygdala). Connections to and from subcortical structures were the focus of the neuroanatomical constraints. The tract-tracing literature provides well-established knowledge on the anatomy of the primate subcortex, with clear homology between human and monkey subcortical structures. This makes it possible to use tract-tracing evidence to refine the connections

intersecting large-scale subcortical structures in human connectivity matrices (we did not consider subdivision of regions into finer nuclei). In contrast, extending tract-tracing evidence to constrain thousands of cortico-cortical links would be intractable and less critical, as most cortico-cortical projections are bidirectional^{37–39}, reducing the risk of false positives due to the lack of directional information. Based on these points, the following neuroanatomical constraints were applied:

- **Cortex ↔ cortex:** All cortico-cortical connections were retained as defined in the original matrices, consistent with evidence that most cortical projections are bidirectional, although their strength may vary with direction^{37–39}.
- **Subcortex → cortex:** Projections to the cortex were retained only for the thalamus^{40,41}, hippocampus^{42,43} and amygdala^{44,45}. Outgoing links from striatal and pallidal regions (caudate, putamen, accumbens and pallidum) to the cortex were excluded, as these projections are unidirectional and originate in the cortex^{21,46,47}.
- **Cortex → subcortex:** With the exception of the pallidum, all subcortical regions in our parcellation receive projections from the cortex, including the striatum^{21,46,47}, thalamus^{40,41}, hippocampus^{42,43} and amygdala^{44,45}. Projections from the cortex to subcortical structures were therefore retained, except those targeting the pallidum⁴⁸.
- **Subcortex ↔ subcortex:** Connectivity within the subcortex was constrained according to established tract-tracing data:
 - **Thalamus:** Sends unidirectional projections to the caudate, putamen and accumbens, receives unidirectional projections from the pallidum and has bidirectional projections with the hippocampus and amygdala^{40,41}.
 - **Caudate and putamen:** Receive unidirectional projections from the thalamus and have bidirectional projections to the pallidum (the FreeSurfer pallidum combines the internal and external globi pallidi)^{21,46,47}.
 - **Pallidum:** Sends unidirectional projections to the thalamus, receives unidirectional projections from the accumbens and amygdala, and exchanges bidirectional projections with the caudate and putamen^{21,48}.
 - **Accumbens:** Sends unidirectional projections to the pallidum and receives unidirectional projections from the thalamus, hippocampus and amygdala^{21,46,47}.
 - **Hippocampus:** Sends unidirectional projections to the accumbens, has bidirectional projections with the amygdala and thalamus, and has bidirectional homotopic projections between the left and right hippocampi^{42,43}.
 - **Amygdala:** Sends unidirectional projections to the pallidum and accumbens, has bidirectional projections to the thalamus and hippocampus, and has bidirectional homotopic projections between the left and right amygdalae^{44,45}.
- **Contralateral connections:** Cortico-subcortical and subcortical-subcortical links were restricted to ipsilateral projections, with the exception of the interhemispheric connections between the left and right hippocampi and amygdalae listed above. This decision is supported by extensive evidence that subcortical circuitry is dominated by ipsilateral connectivity in primates^{46,49,50}. While tracer studies have identified contralateral corticothalamic and cortico-striatal projections, these are sparse and vastly outnumbered by their ipsilateral counterparts^{49,50}.

The neuroanatomical constraints are summarized in the connectivity mask shown in Supplementary Fig. 17. A matrix entry ij indicates the presence (blue) or absence (white) of monosynaptic projections from structure i to structure j . This mask was applied to both normative connectomes via element-wise multiplication, thereby removing likely false-positive links while preserving the tractography-derived weights

of the remaining connections. For the Lausanne connectome, this process removed 1,882 links, yielding a directed connectivity matrix with 2.3% density (a reduction of 0.1% compared with the original 2.4%). For the Schaefer connectome, 8,699 links were removed, resulting in a directed connectivity matrix with 26.58% density (1.3% reduction compared with the original 27.89%). All analyses were carried out using the neuroanatomically constrained connectomes.

Finally, we note that, while these constraints improve the biological validity of the connectomes, they rely on simplifying assumptions and may have drawbacks, such as potential anatomical differences between humans and nonhuman primates. Therefore, for completeness, we replicated key analyses using unconstrained connectomes (Supplementary Fig. 18).

Models of brain network communication

In the main text, we computed brain network communication using shortest path routing. This model identifies the shortest, most direct channel of communication between two brain regions. For a pair of regions that do not share a direct white matter connection—for example, the DLPFC and the SGC—the shortest path comprises a sequence of two or more white matter connections via the connectome. We considered the weighted shortest path, such that the length of a path is quantified by taking into account the total strength of the fibers it comprises.

Formally, let W be the structural connectivity matrix, where entry $W(i,j)$ denotes the strength of the white matter connections between regions i and j . We first defined the matrix of connection lengths $L = -\log(W)$ to quantify the distance or travel cost for pairs of regions. The shortest path between two regions a and b is the sequence of connections $\Pi(a,b) = \{a,u,\dots,v,b\}$ that minimizes the sum $L(a,u) + \dots + L(v,b)$, known as the shortest path length. The capacity for TMS communication from a to b was defined as the number of hops along their shortest path, such that $H(a,b) = |\Pi(a,b)|$. Computed for every pair of regions, this yielded the communication matrix H (Fig. 1a).

In Supplementary Fig. 3, we explored alternative measures of brain network communication that do not quantify communication via shortest paths. The definitions and technical details of these measures can be found in refs. 14,22,51, and the code for their computation is available in the Brain Connectivity Toolbox⁵². Importantly, each measure establishes a clear hypothesis on the direction of association with TMS propagation. For instance, the number of hops along the shortest path quantifies the difficulty or cost of communication, such that paths with more hops are expected to support weaker TMS transmission than paths with fewer hops. Given the clear hypotheses on the directions of associations, we considered one-tailed P values when testing for correlations between network communication measures and treatment response.

Modeling communication from DLPFC TMS sites to the SGC

We computed the average number of white matter hops via the shortest path from TMS sites to the 10-mm-radius SGC sphere centered at MNI (6, 16, −10). We first registered TMS coordinates and gray matter parcellations to MNI space. In some cases, recorded stimulation coordinates were positioned slightly outside the MNI standard brain template and were shifted to the closest within-template gray matter voxel based on Euclidean distance. TMS was modeled to activate gray matter voxels within a 10-mm radius of the stimulation coordinate. Similarly, the SGC sphere comprised gray matter voxels within a 10-mm radius of the SGC coordinate. To ensure analyses captured communication to the SGC, any subcortical voxels in the SGC sphere were discarded. Overlaid onto a gray matter parcellation, each voxel i in the stimulation and SGC spheres was mapped to a parcel $p(i)$ corresponding to a row/column of the structural connectivity matrix.

Communication from patient k 's TMS site to the SGC was computed as a distance-weighted average of the number of hops between voxels

in the stimulation and target spheres. Each voxel i in the stimulation sphere was assigned a weight $w(i) = e^{-\kappa d(i)}$, $\kappa \leq 0$, where $d(i)$ is the Euclidean distance of voxel i from the sphere's center, such that voxels nearer the TMS coordinate contributed more strongly. Voxels in the SGC sphere were uniformly weighted. Communication was then defined as

$$h(k) = \frac{1}{W(k)|T|} \sum_{i \in S(k)} \sum_{j \in T} w(i) H(p(i), p(j)),$$

where $H(p(i), p(j))$ is the number of hops along the shortest path between the parcels of voxels i and j , $S(k)$ is the set of voxels within the stimulation sphere of patient k and $W(k) = \sum_{i \in S(k)} w(i)$.

We note that H is computed on a group-representative connectome, and therefore variations in communication across patients are encoded in $S(k)$ and stem solely from differences in TMS coordinates. In analyses where communication was modeled to loci other than the SGC, the composition of T was adjusted accordingly (for example, Fig. 1d,h). In the analyses of Supplementary Fig. 3, H was replaced by communication matrices from alternative network measures and by functional connectivity matrices. For network measures, we set the decay parameter $\kappa = -1$, reflecting the spatial specificity of white matter pathways, where small variations in the stimulation site may engage different discrete fibers. For functional connectivity, consistent with previous work^{5–7}, we used a uniform weighting ($\kappa = 0$), which better reflects the spatial smoothness of fMRI data. Sensitivity analyses (Supplementary Fig. 19) confirmed that results remained stable across a range of κ values. The communication pathways in Fig. 2 were identified by summing the weighted contributions of paths connecting stimulation and target voxels across patients within each cohort.

Visualization

The cortical maps in Fig. 1d,h were rendered using Cerebro⁵³. To visualize the white matter pathways in Fig. 2g, we first performed tractography on a tissue-unbiased template image of fiber orientation distributions created using multimodal registration⁵⁴. The template was generated in previous work using multimodal registration to guarantee the alignment of white matter tracts with cortical and subcortical gray matter⁵⁴. Streamlines were then transformed to MNI space and visualized using Cerebro⁵³ and Blender. All other figures were created using MATLAB.

Statistics and reproducibility

This study analyzed previously collected clinical and normative brain connectivity datasets and did not involve recruitment of new participants. Datasets are referred to throughout the manuscript as clinical cohort I⁶, clinical cohort II⁷, the Lausanne connectome dataset (discovery)³⁰ and the Schaefer connectome dataset (replication)³⁶. No statistical method was used to predetermine sample size. Sample sizes were determined by the availability of eligible participants in each dataset, as described in the original studies. All computational analyses performed in this study were applied uniformly across participants and datasets and did not involve randomization or blinded allocation of participants. One participant from clinical cohort I was excluded because the recorded TMS stimulation coordinate was positioned outside the MNI brain template, precluding inclusion in the network analyses. No other data were excluded from the analyses. Further information on TMS experimental designs and participant demographics and consent is found above and described in detail in the original studies.

Statistical analyses were performed in MATLAB. Associations between network measures and clinical treatment response were assessed using the Spearman rank correlation coefficient. One-sided tests were used based on a priori hypotheses regarding the direction of association between network communication measures and treatment response. Exact P values were reported where they were numerically representable. Analyses were replicated across four independent

dataset combinations (two clinical cohorts and two normative connectome datasets) to assess robustness. Analysis code and documentation were publicly available to enable the reproducibility of results.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Clinical data (TMS coordinates and rates of symptom improvement) from cohort I are publicly available as part of the Supplementary Information of ref. 6. Clinical data from cohort II (ACTRN12610001071011)⁷ remain subject to privacy and ethical restrictions. The two normative connectome datasets are publicly available through refs. 30,36. The publicly available data used in the analyses of the present manuscript can be found at https://github.com/caioseguin/tms_dep_pathways. Source data are provided with this paper.

Code availability

Code to reproduce the analyses of the present manuscript is available at https://github.com/caioseguin/tms_dep_pathways. Network communication models were computed using the Brain Connectivity Toolbox⁵². Surrogate connectomes were computed using code available through ref. 55.

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Author contributions

Conceptualization: C.S., R.H.F.C. and A.Z. Methodology: C.S., S.M.L., R.F.B. and R.F.H.C. Formal analysis: C.S. Resources: C.S., S.M.L., R.F.B., E.J.C., M.G.P., J.L., L.P., P.B.F. and R.F.H.C. Data curation: R.F.B., P.B.F. and R.F.H.C. Visualizations: C.S. and S.M.L. Writing, original draft: C.S. Writing, review and editing: all authors.

Competing interests

R.F.H.C. and A.Z. are involved in a not-for-profit clinical neuromodulation center (Queensland Neurostimulation Centre) that offers neuroimaging-guided neurotherapeutics. In the last 3 years, P.B.F. has received equipment for research from Brainsway. He has also acted as a founder and board member for TMS Clinics Australia and Resonance Therapeutics. The other authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41593-026-02248-6>.

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Software and code

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Data collection The present study did not collect any data.

Data analysis Data analysis was carried out using MATLAB code (tested on version MATLAB_2021b). Code to reproduce the results of the present manuscript is available at https://github.com/caioseguin/tms_dep_pathways. Network communication models were computed using the Brain Connectivity Toolbox (Rubinov & Sporns, 2010, NeuroImage). Surrogate connectomes were computed using code available through reference (Betzel & Bassett, 2018, PNAS).

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We used data from 4 previous studies:

Clinical cohort I: Weigand A, Horn A, Caballero R, Cooke D, Stern AP, Taylor SF, et al. Prospective Validation That Subgenual Connectivity Predicts Antidepressant Efficacy of Transcranial Magnetic Stimulation Sites. *Biol Psychiatry*. 2018;84: 28–37.

Clinical cohort II: Cash RFH, Zalesky A, Thomson RH, Tian Y, Cocchi L, Fitzgerald PB. Subgenual Functional Connectivity Predicts Antidepressant Treatment Response to Transcranial Magnetic Stimulation: Independent Validation and Evaluation of Personalization. *Biol Psychiatry*. 2019;86: e5–e7.

Normative connectome dataset 1: Griffa A, Alemán-Gómez Y, Hagmann P. Structural and functional connectome from 70 young healthy adults. [cited 29 Jan 2025]. doi:10.5281/zenodo.2872624

Normative connectome dataset 2: Mansour L S, Tian Y, Yeo BTT, Cropley V, Zalesky A. High-resolution connectomic fingerprints: Mapping neural identity and behavior. *Neuroimage*. 2021;229: 117695.

With the exception of clinical data from Cohort II (subject to privacy and ethical restrictions), all other data is publicly available through the references above and at https://github.com/caioseguin/tms_dep_pathways.

Clinical data (TMS coordinates and rates of symptom improvement) from Cohort I is publicly available as part of the Supplementary Information of reference (Weigand et al., 2018, *Biol Psych*). Clinical data from Cohort II (ACTRN12610001071011) (Cash et al., 2021, *Biol Psych*) remain subject to privacy and ethical restrictions. The two normative connectome datasets are publicly available through references (Griffa et al., 2019, Zenodo; Mansour et al., 2021, *NeuroImage*). The publicly available data used in the analyses of the present manuscript can be found at https://github.com/caioseguin/tms_dep_pathways.

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Reporting on sex and gender

The role of sex and gender was not investigated in the present study. Sex demographics for the two clinical cohorts of depression patients were based on self-report and are reported in the manuscript. The term sex was consistently used in the manuscript to indicate the biological attribute.

Reporting on race, ethnicity, or other socially relevant groupings

The present study did not investigate race, ethnicity or other socially relevant groupings.

Population characteristics

Clinical datasets: We consider two independent cohorts of individuals with treatment resistant depression who underwent repetitive TMS to the left DLPFC. Cohort I: N = 25, 17 females, 54.8 ± 9.9 years old (Weigand et al., 2018, *Biol Psych*); Cohort II: N = 24, 10 females, 44.5 ± 11.8 years old (Cash et al., 2021, *Biol Psych*).

Normative datasets: We used two previously published and openly available datasets of normative structural and functional connectivity. The first normative dataset, used in all analyses in the main text, comprised brain networks for 70 healthy young adults (mean age 28.8 ± 9.1, 27 women) acquired at the University of Lausanne, Switzerland (Griffa et al., 2019, Zenodo). The second dataset comprised brain networks from 1,000 Human Connectome Project (HCP) healthy young adults (Mansour et al., 2021, *NeuroImage*; Van Essen et al., 2012, *NeuroImage*).

Recruitment

The present study did not recruit participants. Procedures for recruitment are detailed in:

Clinical cohort I: Weigand A, Horn A, Caballero R, Cooke D, Stern AP, Taylor SF, et al. Prospective Validation That Subgenual Connectivity Predicts Antidepressant Efficacy of Transcranial Magnetic Stimulation Sites. *Biol Psychiatry*. 2018;84: 28–37.

Clinical cohort II: Clinical trial identifier ACTRN12610001071011; Cash RFH, Zalesky A, Thomson RH, Tian Y, Cocchi L, Fitzgerald PB. Subgenual Functional Connectivity Predicts Antidepressant Treatment Response to Transcranial Magnetic Stimulation: Independent Validation and Evaluation of Personalization. *Biol Psychiatry*. 2019;86: e5–e7

Normative connectome dataset 1: Griffa A, Alemán-Gómez Y, Hagmann P. Structural and functional connectome from 70 young healthy adults. [cited 29 Jan 2025]. doi:10.5281/zenodo.2872624

Normative connectome dataset 2 (Human Connectome Project): Van Essen, David C., et al. "The Human Connectome Project: a data acquisition perspective." *Neuroimage* 62.4 (2012): 2222–2231.

Ethics oversight

Institutional research ethics committees approved the studies of each data collection site.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	The present study did not recruit participants. Procedures for recruitment are detailed in: Clinical cohort I: Weigand A, Horn A, Caballero R, Cooke D, Stern AP, Taylor SF, et al. Prospective Validation That Subgenual Connectivity Predicts Antidepressant Efficacy of Transcranial Magnetic Stimulation Sites. <i>Biol Psychiatry</i>. 2018;84: 28–37. Clinical cohort II: Clinical trial registry ACTRN12610001071011; Cash RFH, Zalesky A, Thomson RH, Tian Y, Cocchi L, Fitzgerald PB. Subgenual Functional Connectivity Predicts Antidepressant Treatment Response to Transcranial Magnetic Stimulation: Independent Validation and Evaluation of Personalization. <i>Biol Psychiatry</i>. 2019;86: e5–e7 Normative connectome dataset 1: Griffa A, Alemán-Gómez Y, Hagmann P. Structural and functional connectome from 70 young healthy adults. [cited 29 Jan 2025]. doi:10.5281/zenodo.2872624 Normative connectome dataset 2 (Human Connectome Project): Van Essen, David C., et al. "The Human Connectome Project: a data acquisition perspective." <i>Neuroimage</i> 62.4 (2012): 2222-2231.
Data exclusions	Clinical cohort II (clinical trial registry ACTRN12610001071011; Cash et al., 2021, <i>Biol Psych</i>) comprises 25 participants with depression who underwent TMS treatment to the left DLPFC for 5-8 weeks. One participant was excluded due to the recorded TMS coordinate being positioned outside of the MNI brain template. Analyses involving this cohort were carried out using data from the remaining 24 participants.
Replication	Analyses were replicated in a total of 4 combinations of independent datasets: 2 clinical cohorts and 2 normative connectome datasets, as described above.
Randomization	N/A. All patients with depression in both clinical samples received the same TMS protocol for the duration of the TMS therapy pertaining to the analyses. Data from all healthy participants in normative datasets were included in the analyses.
Blinding	Blinding is not applicable here. As described above, none of the participants (in neither clinical nor healthy samples) were allocated to experimental groups.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☐	☒ MRI-based neuroimaging

Plants

Seed stocks	Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.
Novel plant genotypes	Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.
Authentication	Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosaicism, off-target gene editing) were examined.

Magnetic resonance imaging

Experimental design

Design type

Structural, diffusion and functional MRI

Design specifications

We did not collect new MRI data for this study.

We used template structural MRI volumes in 2mm MNI152 of (i) FSL standard T1-weighted brain image, (ii) the Lausanne parcellation (scale 5) and (iii) the Schaefer parcellation (800 regions, 7 Yeo networks). Cortical parcellations had been previously registered to MNI152 in Griffa et al., 2019, Zenodo (Lausanne parcellation) and Mansour et al., 2021, NeuroImage (Schaefer parcellation). These volumes were used to determine which gray matter regions were activated by TMS delivered to specific coordinates, according to the strategy described in the present manuscript's methods and implemented in https://github.com/caioseguin/tms_dep_pathways.

We used two previously published and openly available datasets of normative structural and functional connectivity. We considered connectivity data that was already processed into region-by-region connectivity matrices, and did not carry out any brain imaging acquisition or preprocessing. Details on the two datasets are provided below.

The first normative dataset, used in all analyses in the main text, comprised brain networks for 70 healthy young adults (mean age 28.8 ± 9.1 , 27 women) acquired at the University of Lausanne, Switzerland (Griffa et al., 2019, Zenodo). Grey matter was parcellated into 1,000 cortical and 14 subcortical (left and right thalamus, caudate, putamen, pallidum, accumbens area, hippocampus and amygdala) regions. Structural connectivity was estimated from the diffusion MRI of individual participants using deterministic white matter tractography. The strength of the white matter connection between two regions was computed as the number of streamlines intersecting them divided by the average of their surface areas. Resting-state fMRI volumes underwent preprocessing to correct for motion, physiological confounds and global signal. The functional connectivity of each individual was computed as the Pearson correlation between regional time courses of blood oxygenation level dependent (BOLD) signals. Further details on image acquisition and connectivity reconstruction are found in (Griffa et al., 2019, Zenodo).

The second dataset, used in the replication analysis of Fig S3, comprised brain networks from 1,000 Human Connectome Project (HCP) healthy young adults. Details on HCP imaging acquisition and pre-processing are described in (Van Essen et al., 2012, NeuroImage). Grey matter regions were defined based on the on the Schaefer 800 (7 Networks) parcellation combined with the same 14 subcortical regions listed above. Structural connectivity was computed using probabilistic white matter tractography applied to diffusion MRI, and the strength of the structural connections was inferred as the streamline count between grey matter regions. Functional connectivity was once again computed as the Pearson correlation of regional BOLD time courses following resting-state fMRI preprocessing. Further details on connectivity reconstruction are provided in (Mansour et al., 2021, NeuroImage).

Behavioral performance measures

NA

Acquisition

Imaging type(s)

NA

Field strength

NA

Sequence & imaging parameters

NA

Area of acquisition

NA

Diffusion MRI

☐

Used

☒

Not used

Preprocessing

Preprocessing software

NA

Normalization

Na

Normalization template

NA

Noise and artifact removal

NA

Volume censoring

NA

Statistical modeling & inference

Model type and settings

We did not perform statistical inference at the level of brain images. Instead, we considered network (i.e., graph-theoretical) measures computed from DLPFC TMS coordinates to the subgenual cingulate cortex seed (10-mm sphere centred at MNI [6,16,-10]). This yielded a single value of DLPFC-SGC white-matter communication per patient. Network communication

measures were correlated to treatment response (symptom improvement) following TMS therapy. Refer to the Graph analysis section below for more details.

Effect(s) tested

The Spearman rank correlation coefficient (one-sided test) was used to test for a relationship between network measures and treatment response. The Spearman rank correlation was chosen because (i) network measures do not always meet the assumptions required for parametric statistical tests (e.g., normality) and (ii) it is not limited to capturing linear effects. One-sided tests were considered because of clear hypotheses on the direction of association between network measures and treatment response (see Methods for details).

Specify type of analysis: ☒ Whole brain ☐ ROI-based ☐ Both

Statistic type for inference

NA (see above).

(See [Eklund et al. 2016](#))

Correction

Benjamini-Hochberg FDR method

Models & analysis

n/a Involved in the study

- ☒ ☐ Functional and/or effective connectivity
- ☐ ☒ Graph analysis
- ☒ ☐ Multivariate modeling or predictive analysis

Graph analysis

We applied an established consensus method to generate group-representative connectomes from the structural connectivity matrices of individual participants (Betzel et al., 2019, *Netw Neurosci*). This method selects the white matter connections most reliably found across individuals, while preserving the sample's average connection density, proportions of intra- and inter-hemispheric connections, and distribution of connection lengths. The resulting consensus matrices yield prototypical maps of human brain connectivity that mitigate tractography reconstruction errors, filter out individual-level artefacts, and boost signal-to-noise ratio. For the Lausanne dataset, this yielded a $1,014 \times 1,014$ structural connectivity matrix with 2.4% connection density; for the Schaefer dataset, an 814×814 matrix with 27.9% connection density.

To improve the biological validity of these connectomes, we applied a neuroanatomical constraint procedure based on tract-tracing evidence from non-human primates. This preprocessing step systematically and conservatively removed subcortical and cortico-subcortical connections that are likely to represent tractography false positives (e.g., implausible bidirectional projections between the cortex and pallidum, or contralateral striatal crossings), while retaining established monosynaptic projections consistent with primate anatomy. After this refinement, the Lausanne connectome had a density of 2.3% and the Schaefer connectome 26.6%. All graph analyses were performed on these neuroanatomically constrained connectomes.

We computed brain network communication using the weighted shortest path routing model, which identifies the most direct polysynaptic route between two brain regions. For pairs of regions without a direct white matter connection—such as the dorsolateral prefrontal cortex (DLPFC) and the subgenual cingulate cortex (SGC)—the model quantified the number of white matter “hops” through the connectome. For each patient, TMS stimulation and SGC target sites were modelled as 10-mm spheres in grey matter space, and the average number of hops between them was computed, with distance weighting reflecting spatial proximity to the stimulation coordinate. Variability in communication across patients arose solely from differences in stimulation coordinates, as connectomes were group-representative.

All analyses were repeated across both normative connectomes, with consistent results. Functional connectivity matrices, used in comparative analyses (e.g., Fig 3 and Fig S3), were obtained by averaging participant-level functional connectivity across each dataset.